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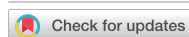
Evaluating a landscape-scale model to forecast wheat stem rust

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Evaluating a landscape-scale model to forecast wheat stem rust

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Abstract

Wheat stem rust is a fungal disease that can cause total loss of a wheat crop. Outbreaks are typically far-reaching because infectious spores can disperse aerially over thousands of kilometres. Rapid response to an outbreak requires a timely and accurate estimate of disease risk. We constructed an integrated mechanistic model of stem rust infection driven by seven-day meteorological forecasts and near-real time disease surveillance to inform mitigation at the national scale. Focusing on Ethiopia as the largest wheat producer of sub-Saharan Africa, yet where food insecurity is high, we evaluate the predictability of wheat stem rust infections during the rain-fed seasons of 2015–2022. The epidemiological model performs well at predicting disease observations two weeks ahead of time, from cross-validation root-mean-square-error values were in the range 0.25–0.35, ROC skill score 0.58–0.68, and odds ratio skill score 0.78–0.90. Results of alternative model scenarios show the benefit of assimilating recent surveys and simulating spore dispersal to achieve a good prediction. When driven by seven-day weather forecasts, the model therefore provides up to 21 days to respond to likely stem rust infections and prevent epidemic spread and subsequent crop loss. The mechanistic form and large scale of the epidemiological model has utility beyond in-season predictions, particularly with respect to scenario testing for practical interventions. The results demonstrated that long-distance dispersal has an important role in the spread of stem rust in a region of diverse agro-ecology. The forecast system presented here can be adapted to other plant pathosystems where aerial dispersal is the primary risk of disease spread.

1. Introduction

Fungal diseases caused by wheat rust pathogens are major biotic constraints for worldwide wheat production (Savary *et al* 2019). Wheat stem rust, caused by *Puccinia graminis* f. sp. *tritici*, can completely destroy susceptible crops (Roelfs *et al* 1992, Olivera *et al* 2015), and new emerging virulent strains further threaten food security (Bhavani *et al* 2022). The scale of wheat stem rust outbreaks is typically large because infectious urediniospores (hereafter ‘spores’)

can disperse aerially over hundreds to thousands of kilometres (Brown and Hovmöller 2002). Mitigating disease impact requires early detection and adaptation (Ristaino *et al* 2021, Morris *et al* 2022).

Disease models can support adaptation in the form of risk forecasts and scenario simulations, for example to compare the likely effectiveness of management strategies. However, to be useful at the landscape scale, a model must be parameterised and validated for which there is a number of challenges (Cunniffe *et al* 2015) relating to spatial

heterogeneity, interannual variability, and pathogen dispersal. Wheat rust models have so far been developed for limited use cases. For instance, simple statistical models based on multi-year averages of observations identify recurring hotspots of infection but miss interannual variability (Meyer *et al* 2021). Most mechanistic wheat disease models to date have focused on continuous observations at a limited set of locations thereby neglecting long-distance dispersal (Salotti *et al* 2022, González-Domínguez *et al* 2023). Meanwhile, attempts have been made to account for long-distance dispersal in other pathosystems sensitive to airborne inoculum (Isard *et al* 2007, Radici *et al* 2024).

Ethiopia's food security is affected by wheat rusts in a complex host landscape. It is the largest producer of wheat in sub-Saharan Africa (FAOSTAT 2025) with multiple, overlapping, wheat cropping periods. Large wheat rust epidemics have occurred, with severe impacts on yields in 1988, 1993–94, 2010, 2013–2014, and 2016 (Olivera *et al* 2015, Jaleta *et al* 2019, Meyer *et al* 2021). Recently, 59% of Ethiopia's population is reported to have experienced moderate or severe food insecurity during 2021–2023 (table A1.1 in The State of Food Security and Nutrition in the World 2024—*Financing to end hunger, food insecurity and malnutrition in all its forms*, 2024), and there is a national food security policy to expand wheat production (Effa *et al* 2023). Surveillance for wheat rust has been conducted in Ethiopia since 2007 (RustTracker 2023), involving ~1000 field surveys annually. Amongst wheat rusts, the prevalence of stem rust and stripe rust (caused by *P. striiformis*) has been generally high with substantial annual variability (Meyer *et al* 2021).

Since 2018, a wheat rust early warning system (EWS) for Ethiopia has been successfully producing daily risk warnings known as 'wheat rust advisories' that have reached more than 300 000 farmers annually (Allen-Sader *et al* 2019). Incorporating near real-time field surveillance, host distribution maps and numerical weather prediction output, the EWS forecasts the rate of spore deposition and the environmental suitability for spore germination over the next seven days. Farmers then have up to 21 d to respond to warnings of likely stem rust infection by scouting fields and applying fungicide if necessary to prevent local epidemic spread and subsequent crop loss. An early qualitative validation showed close correspondence between spore dispersal forecasts and infections sighted in a susceptible variety during a pilot trial of the forecast system (Olivera *et al* 2015). Since then, the numbers of district advisers and farmers accessing the outputs has grown steadily, with requests and ongoing implementation in seven more countries in East Africa and South Asia, providing support for the value of the EWS. However, formal evaluation has

until now not been conducted, in part, because of a disjoint between model output, which relates to spore arrival and infection and evaluation data that relate to expression of disease symptoms.

Here we solve the problem by integrating a landscape-scale disease model for stem rust with the dispersal and infection suitability components of the EWS to enable comparison between model predictions and field observation of disease. The primary objectives of the study are to parameterise and validate the integrated epidemiological model with field surveys, and to evaluate how model components affect overall predictability.

2. Data, models, and methods

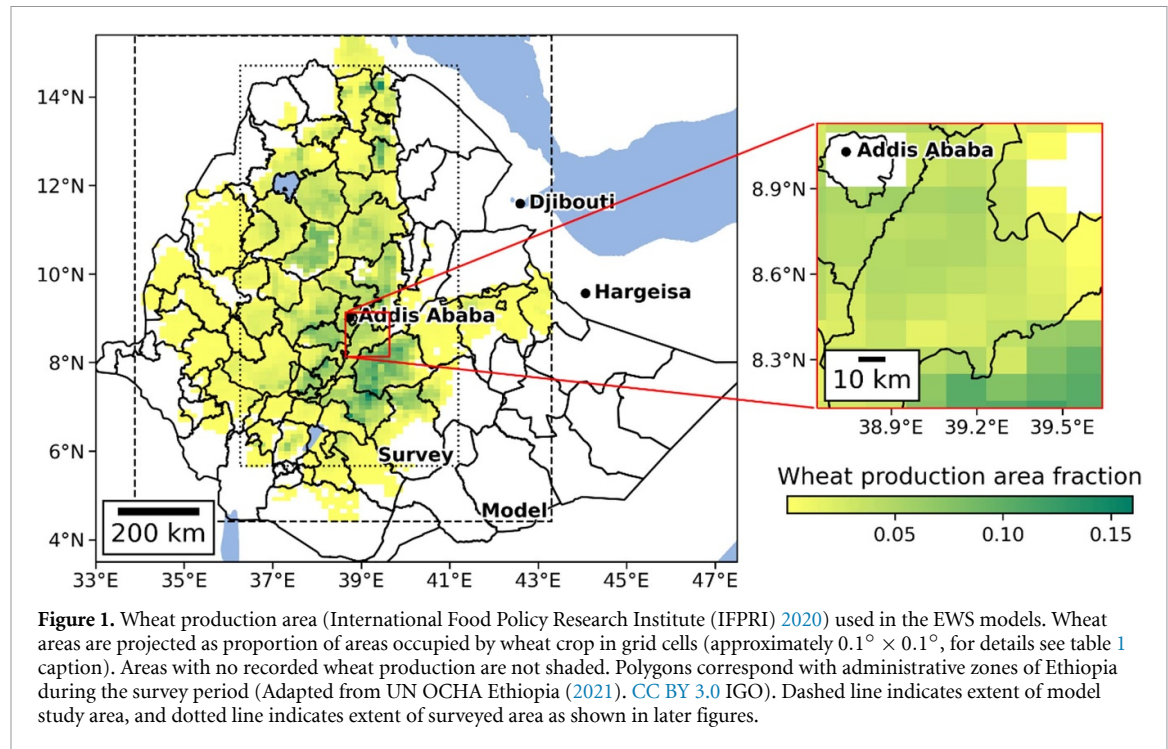
2.1. Study area and host distribution

We focus on the susceptible period of rain-fed wheat production in Ethiopia (the short-rain, *Belg*, season; and the main, *Meher*, season), 15th July–30th December each year. We further restrict our study to the years 2015–2022, corresponding with the availability of a static crop map for wheat cultivation in Ethiopia (International Food Policy Research Institute (IFPRI) 2020) which is used as a base upon which dispersal and subsequent infection are simulated (see figure 1). The study area spans from 4.4° N, 33.9° E, to 15.4° N, 43.3° E (dashed line in figure 1).

2.2. Field surveys

Field surveys of wheat rust have been conducted across Ethiopia since 2007 (RustTracker 2023). Surveyed fields were randomly selected at semi-regular intervals of several kilometres along transects (following roads). Surveys were conducted by trained plant pathologists using an international standard protocol (Ali and Hodson 2017). The surveyors recorded within-field stem rust incidence (categorised as none; low: 0%–20% plants infected; moderate: 20%–40%; or high: >40%), average severity (0%–100%) using the modified Cobb scale, and plant reaction (Peterson *et al* 1948, Roelfs *et al* 1992) in each surveyed field.

Field surveillance presented several challenges for model fitting and testing. The date of disease onset was never observed directly because individual sites are visited only once. Due to the extensive area under wheat production and despite intensive surveillance efforts, survey coverage was sparse (figure 2(a)). For example, in the rain-fed seasons of 2019, the 973 surveys conducted over roughly 33 administrative zones covered approximately 1445 ha (figure 2(b)), or 0.1% of the estimated 1.8 M ha harvested area (FAOSTAT 2025). Furthermore, coverage varied spatially between years (figure 2(a)). Coverage also varied temporally, as indicated by the somewhat step-wise nature of the surveyed area in figure 2(b). Off-season



surveillance changed since 2019, reflecting increases in areas of irrigated wheat and COVID-related restrictions applied in early 2021.

To address the differences between the coarse temporal and spatial resolution of surveys and the regular gridded model outputs, we calculated the average disease levels for survey and model outputs in a series of spatiotemporal zones (see figure 2(c)). Averaging within each zone provided consistent units for both survey data and model predictions, enabling model parameterisation and prediction. The default structure was based on administrative zones (cf figure 1(a)) divided into 14 d windows.

2.3. Models of the wheat rust EWS

The wheat rust EWS comprises a sequence of models to compute source strength, spore dispersal, deposition and infection at deposition sites (Allen-Sader *et al* 2019). Spore availability and source strength are calculated from field surveys in near-real time. These data represent the best knowledge of the state of infection across the landscape. Spore availability and source strength, D_{SOURCE} , are calculated at source locations, $\mathbf{x}_s$, and times, t_s , across the landscape:

$$D_{\text{SOURCE}}(\mathbf{x}_s, t_s) = F(I_0(\mathbf{x}_s, t_s)), \quad (1)$$

where F represents local (within grid cell) spore production estimated from average infection levels I_0 (obtained from field surveys) within model grid cells (for details, see Allen-Sader *et al* 2019).

Long-distance dispersal of spores from source locations is simulated by the particle dispersion model NAME (Jones *et al* 2007) adapted for wheat rust

spores (Meyer *et al* 2017b, Allen-Sader *et al* 2019, Gilligan 2024). Spore deposition rates, $D_{\text{NAME}}(\mathbf{x}, t)$, within each grid cell, are described as:

$$D_{\text{NAME}}(\mathbf{x}, t) = G(D_{\text{SOURCE}}(\mathbf{x}_s, t_s), \mathbf{x}, t), \quad (2)$$

where G maps spore quantities from all source locations, $\mathbf{x}_s$, and prior release times, t_s , to deposited locations, $(\mathbf{x}, t)$, according to biological and meteorological processes including canopy escape, turbulent transport, decay due to exposure to ultraviolet light, and deposition. A subsequent model calculates environmental suitability, $E(\mathbf{x}, t)$, for spore infection from 2 m altitude air temperature and an estimated dew period from 2 m altitude relative humidity and precipitation (Allen-Sader *et al* 2019). For the purposes of the current paper, the models for spore dispersal and environmental suitability for infection were driven by meteorology from the UK Met Office global analysis dataset (Met Office 2013, 2024) with a time step of three hours.

2.4. Integrated epidemiological model of wheat stem rust

We define an epidemiological model with two host compartments, susceptible S and infected I . At the start of the season, $t = t_0$, it is assumed there is no infected host. The area fraction of susceptible and infected host per grid cell (with units of $m^2 m_{\text{grid cell}}^{-2}$) is fixed by the area of wheat present in a grid cell, $H(\mathbf{x})$, as shown in figure 1, so that:

$$H(\mathbf{x}) = S(\mathbf{x}, t_0) = S(\mathbf{x}, t) + I(\mathbf{x}, t). \quad (3)$$

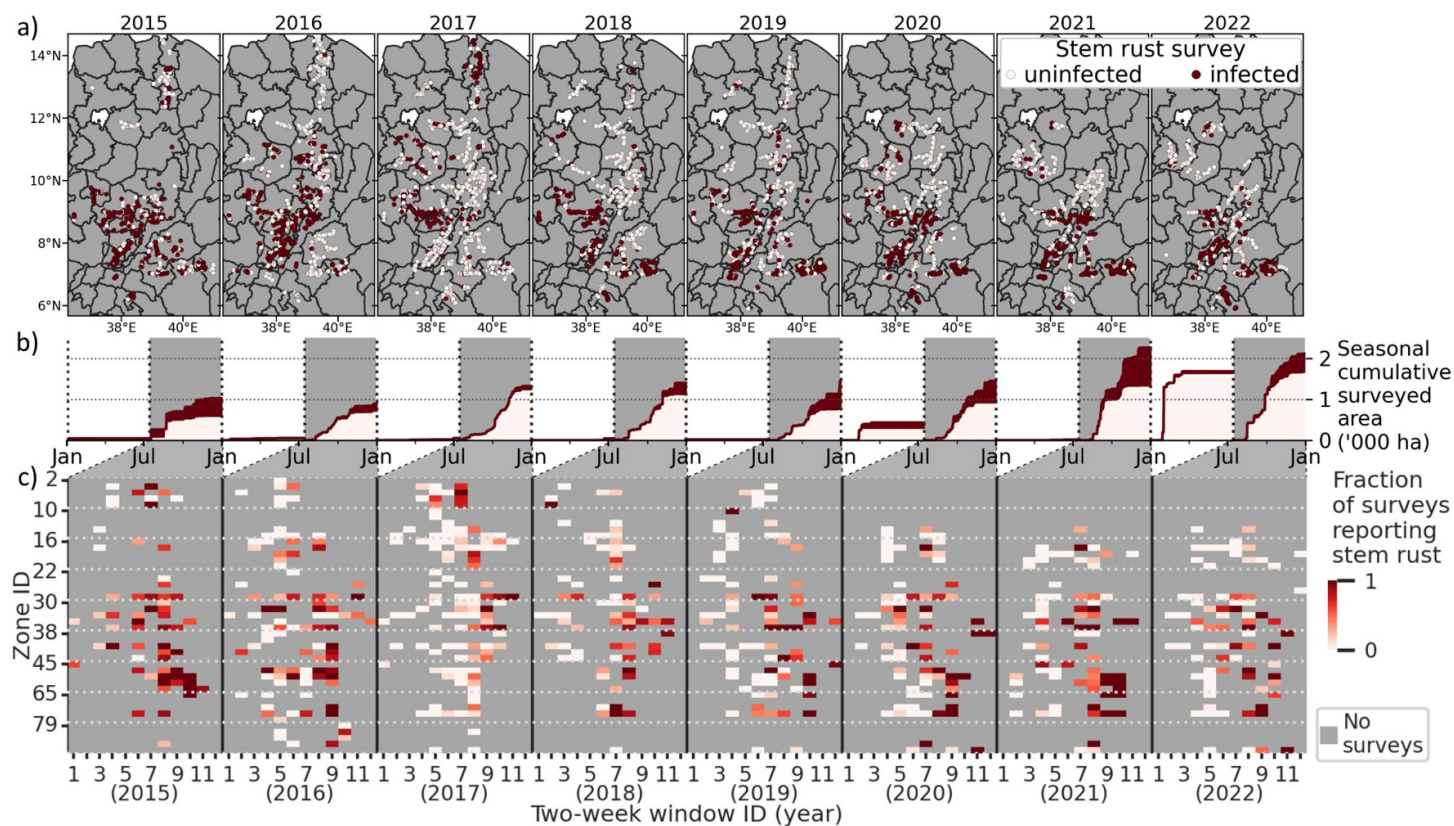
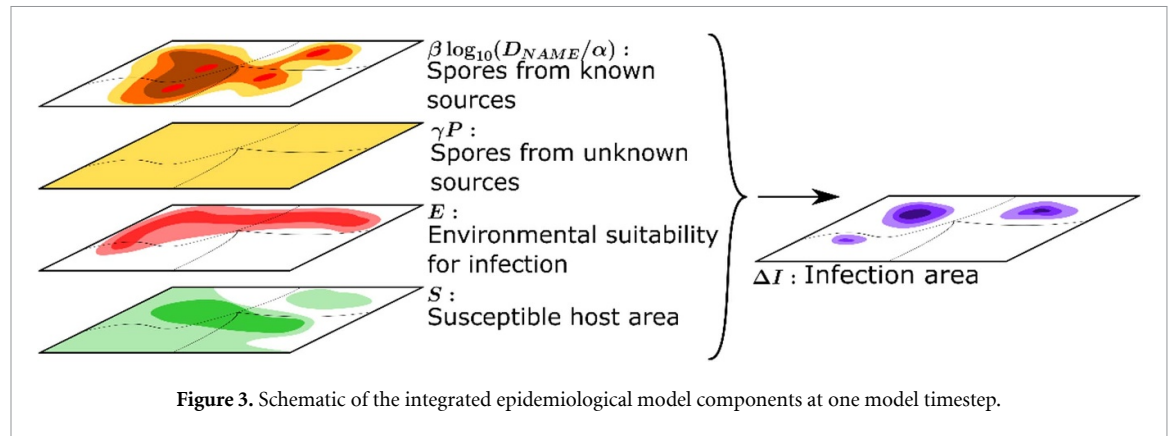


Figure 2. Wheat rust survey distributions. (a) Stem rust survey locations and infection presence in rain-fed wheat seasons during 2015–2022. (b) Cumulative area surveyed in Ethiopia for wheat stem rust each year between 2015 and 2022, shading indicates period within rain-fed seasons that is simulated (15th July–30th December). (c) Fraction of surveys that reported stem rust infection in each administrative zone and each two-week time window during the simulated periods. Correspondence between survey dates and simulated periods is shown between (b) and (c). Unsurveyed zones (47 of the 92 total) are not shown here but can be found in supplementary figure S2. The map of administrative zone IDs is shown in supplementary figure S1.



The change in the area fraction of infected host, I , within each grid cell over time, t , is formulated as:

$$\frac{dI(\mathbf{x}, t)}{dt} = S(\mathbf{x}, t) f(\mathbf{x}, t), \quad (4)$$

where f is the viable spore challenge rate over the grid cell—the number of attempts per second by spores that could successfully lead to infection within that cell. Alternative structures for f are considered later, but we focus on:

$$f(\mathbf{x}, t) = \begin{cases} C(\mathbf{x}, t) E(\mathbf{x}, t) [\beta \log_{10}(\alpha^{-1} D_{\text{NAME}}(\mathbf{x}, t)) + \gamma P(\mathbf{x}, t)], & \text{where } D_{\text{NAME}}(\mathbf{x}, t) > \alpha \\ C(\mathbf{x}, t) E(\mathbf{x}, t) \gamma P(\mathbf{x}, t), & \text{otherwise,} \end{cases} \quad (5)$$

where, $C(\mathbf{x}, t)$ is a function that modulates spore viability at deposition sites by farmer control, $E(\mathbf{x}, t)$ modulates environmental (meteorological) conditions for infection. Following a pilot study, the daily maximum of environmental suitability is used. $P(\mathbf{x}, t)$ accounts for ingress of spores from unknown sources not otherwise accounted for by the model calculations. In the absence of additional knowledge and to simplify the analysis, we assume a spatially and temporally uniform challenge from unknown sources of inoculum, setting P to 1. The function $C(\mathbf{x}, t)$ is introduced to add flexibility for future work incorporating control interventions. During model fitting and testing, we assumed no variation in control practice across the landscape within the study period by setting $C(\mathbf{x}, t) = 1$. The component contributions to the integrated epidemiological model are illustrated in figure 3.

The two terms in square brackets in equation (5) represent spore challenge rates from known and unknown sources of inoculum. Unknown sources include spore dispersal from unsurveyed regions (grey areas in figure 2(c)), and possibly incursion from beyond Ethiopia) and potentially localised aeciospore spread from the alternate host, barberry. Their respective weightings are set by the parameters β and γ : we explore values ranging from zero (no influence of spores from known and unknown sources) to large values that facilitate extremely rapid epidemic development (table 1). A third parameter, α , is necessary

to non-dimensionalise the logarithm. Furthermore, since the particle dispersion model calculates millions of trajectories to represent the probability of release and aerial dispersal of many ($\gg$ trillions of) spores, the lowest rates of spore deposition may correspond to extremely low probabilities. Including the lowest rates of spore deposition may introduce unnecessary noise in the predictions (Aylor 2003, Isard *et al* 2007, Meyer *et al* 2017b). We consider exclusion thresholds for low deposition rates, α , ranging from 0.0001 to 100 spores $\text{m}^{-2} [3 \text{ h}]^{-1}$ (table 1).

We assume a period of 14 d between infection and detectable symptoms in field surveys. The incubation period between spore arrival and infection and symptom expression varies with temperature, with a minimum period of six days in optimum conditions (Nopsa and Pfender 2014). We do not simulate the incubation period because it occurs in—and typically extends beyond—the seven-day forecast period of the EWS. The results of Nopsa and Pfender (2014) indicate that 14 d is sufficient for incubation to have occurred across a broad range of temperatures (12.1 °C–30.4 °C) typical of Ethiopia's wheat production areas in the rain-fed seasons. Therefore, predictions of the state of infection are compared with observations 14 d in the future.

The time step of the epidemiological model is three hours. The model's spatial grid is set to match the underlying meteorological model, which changed in early 2017 from $0.234375^\circ \times 0.15625^\circ$

Table 1. The principal variables and parameters of the integrated epidemiological model.

Notation	Units	Definition	Source
S	$\text{m}^2 \text{m}_{\text{grid cell}}^{-2}$	Susceptible area fraction. Range: 0–1	Initial state: MapSPAM 2017 SSA (International Food Policy Research Institute (IFPRI) 2020)
I	$\text{m}^2 \text{m}_{\text{grid cell}}^{-2}$	Infected area fraction. Range: 0–1	—
f	$[\text{3 h}]^{-1}$	Rate of infection	—
C	1	Control factors affecting infection efficiency, e.g., fungicide use	—
E	1	Efficiency of infection due to environmental factors. Range: 0–1	Met Office UM (Walters <i>et al</i> 2019) and environmental suitability model
D_{NAME}	$\text{spores m}^{-2} [\text{3 h}]^{-1}$	Spore deposition rate from known infections	Met Office UM and NAME (Jones <i>et al</i> 2007)
P	$\text{spores m}^{-2} [\text{3 h}]^{-1}$	Spore deposition rate from unknown infections	—
α	$\text{spores m}^{-2} [\text{3 h}]^{-1}$	Lower limit of spore deposition rate from known infections	Parameterisation based on survey results. Values calculated: 0.0001, 0.01, 1, 100.
β	$[\text{3 h}]^{-1}$	Rate of infection by known sources generated by the model	Parameterisation based on survey results. 0.0, and 16 values spaced evenly logarithmically between 0.0001 and 0.1.
γ	m^2/spore	Infected area per spore from unknown sources not generated by the model	Parameterisation based on survey results. 0.0, and 11 values spaced evenly logarithmically between 0.0001 and 0.01.

(roughly 25×16 km at the latitude of Ethiopia) to $0.140625^\circ \times 0.09375^\circ$ (roughly 16×10 km). The principal variables and parameters for the integrated epidemiological model are listed in table 1.

2.5. Parameterising the epidemiological model

To estimate the three parameters, α , β , and γ in the epidemiological model, we ran the model over a global search of all parameter combinations (listed in table 1) and determined the best fit using a combination of three metrics which compare two variables (summarised in table 2): root-mean-square-error (RMSE), receiver-operator characteristic skill score (RSS) and odds ratio skill score (ORSS). The choice of metrics was motivated in part to consider epidemic behaviour at different spatial scales: broad administrative zones (RMSE) and site-by-site (RSS and ORSS). Other metrics were calculated (e.g., $F1$ score) but, after a pilot study, were excluded from the fitting procedure as they were found to exhibit similar characteristics in parameter space to RMSE, RSS or ORSS.

To calculate RMSE, the training surveys and predictions were first averaged within each administrative zone and each two-week window, as shown in figure 2(c) for surveys. The values of RMSE can range

from 0 to 1, where zero indicates perfect agreement. The RSS compared the predicted area infected at observation sites against reported presence of infection. Thirdly, ORSS also involved comparisons at the scale of observation sites, but instead used the categorised model predictions (i.e., infected or uninfected) for each survey site. We followed Meyer *et al* (2021) and categorised the model predictions from predicted infected area using the threshold that results in the highest number of true positives and true negatives (maximum of Youden's index). The RSS is therefore a measure of overall model performance and the ORSS is a separate indicator focused on how site-specific disease presence or absence is interpreted from grid cell predictions (Nurmi 2003). We note that site-specific predictions were drawn from the values at the model grid scale (~ 10 km) and so sites in the same grid cell were assigned the same prediction. The values of the RSS and ORSS range from 0 to 1, where 0 indicates no skill and 1 indicates perfect skill.

The optimum parameterisation was defined as the point in parameter space with the smallest total difference from the best realised values of all three metrics. That is, for a metric, X , the weighted difference is defined as:

Table 2. Description of metrics used in parameterisation and evaluation of model against wheat rust surveys.

Metric	Abbreviation	Theoretical worst value or no skill	Theoretical best value	Observation format	Prediction format
Root mean square error	RMSE	1	0	Incidence fraction per administrative zone (figure 2(c))	Infected area fraction I per administrative zone
Receiver-operator characteristic (ROC) skill score	RSS	0	1	Infection presence/absence at survey site (figure 2(a))	Infected area fraction I in model grid cell
Odds ratio skill score	ORSS	0	1	Infection presence/absence at survey site (figure 2(a))	Infection presence/absence in model grid cell (see text)

$$E_X(\alpha, \beta, \gamma) = \frac{X_{\text{best}} - X(\alpha, \beta, \gamma)}{X_{\text{perfect}} - X_{\text{no-skill}}}, \quad (6)$$

where X_{best} is the best realised values across the range of computed parameters, X_{perfect} is the best theoretical value, and $X_{\text{no-skill}}$ is the worst theoretical value for the metric or, in the case of skill scores, it is the value when the metric shows no skill. Difference increases as E_X increases from 0 to 1. The overall fit is the maximum of these weighted differences:

$$E_{\text{overall}}(\alpha, \beta, \gamma) = \max(E_{\text{RMSE}}(\alpha, \beta, \gamma), E_{\text{RSS}}(\alpha, \beta, \gamma), E_{\text{ORSS}}(\alpha, \beta, \gamma)). \quad (7)$$

The best fit is where $E_{\text{overall}}(\alpha, \beta, \gamma)$ is smallest. We tested various approaches to minimising this difference (e.g., by percentile or metric units) and reached consistently similar results when training over multiple years.

2.6. Evaluating performance

Training and testing were conducted by separating data into four-year periods within 2015–2022. We expected considerable interannual variability due to annual changes in survey coverage and shifting farmer preferences for cultivated varieties. Therefore, we conducted two-fold cross-validation with shuffling of annually grouped surveys. Fitting and testing was conducted on every possible combination of the eight-year period split into two groups of four years, of which there are 70. Cross-validation with different durations of training and testing (for example, four-fold: six training years and two test years) yielded similar results to comparisons based upon four years (see supplementary figures S1 and S2). The same metrics as used in the fitting procedure were used to evaluate model performance in the test years.

2.7. Evaluating alternative model scenarios

The sequence of models to compute source strength, spore dispersal, deposition and infection at deposition sites, together with the integrated epidemiological model are defined as a baseline model against which several simplified or adapted variants of the model were compared. We tested scenarios that differed in which model components were incorporated, as well as for which survey data were included. For each scenario, we fitted the model and cross-validated performance as described in sections 2.5 and 2.6. The difference in performance from the baseline setup was measured for each four-year test combination arising from cross-validation. We considered the performance of a scenario to be similar to the baseline if the interquartile range of differences of all test year combinations overlapped with zero.

3. Results

3.1. Parameterisation

An example of goodness-of-fit over parameter space is shown in figure 4 for the training years 2015–2018 for each of the three metrics. While there is broad consistency amongst the metrics, for example in low scores for configurations of spore challenge that are extremely weak (β and γ close to zero) or extremely strong (large β and γ), the estimated values of β and γ for optimal fits differ for each metric. The RMSE indicates a trade-off between β and γ to reach the best fit, the two parameters compensate for each other, while the RSS and ORSS metrics indicate that a smaller γ and larger β lead to an improved fit between the model predictions and field observations. The optimal parameter combination based on all three metrics is outlined as a green square in figure 4.

Across the possible combinations of four-year training periods, best fits were typically between $\beta = 0.001$ and 0.01, around $\gamma = 0$ and 0.0001, and $\alpha = 1$ and 100. These values indicate the weightings for spore dispersal (β) varied with the choice

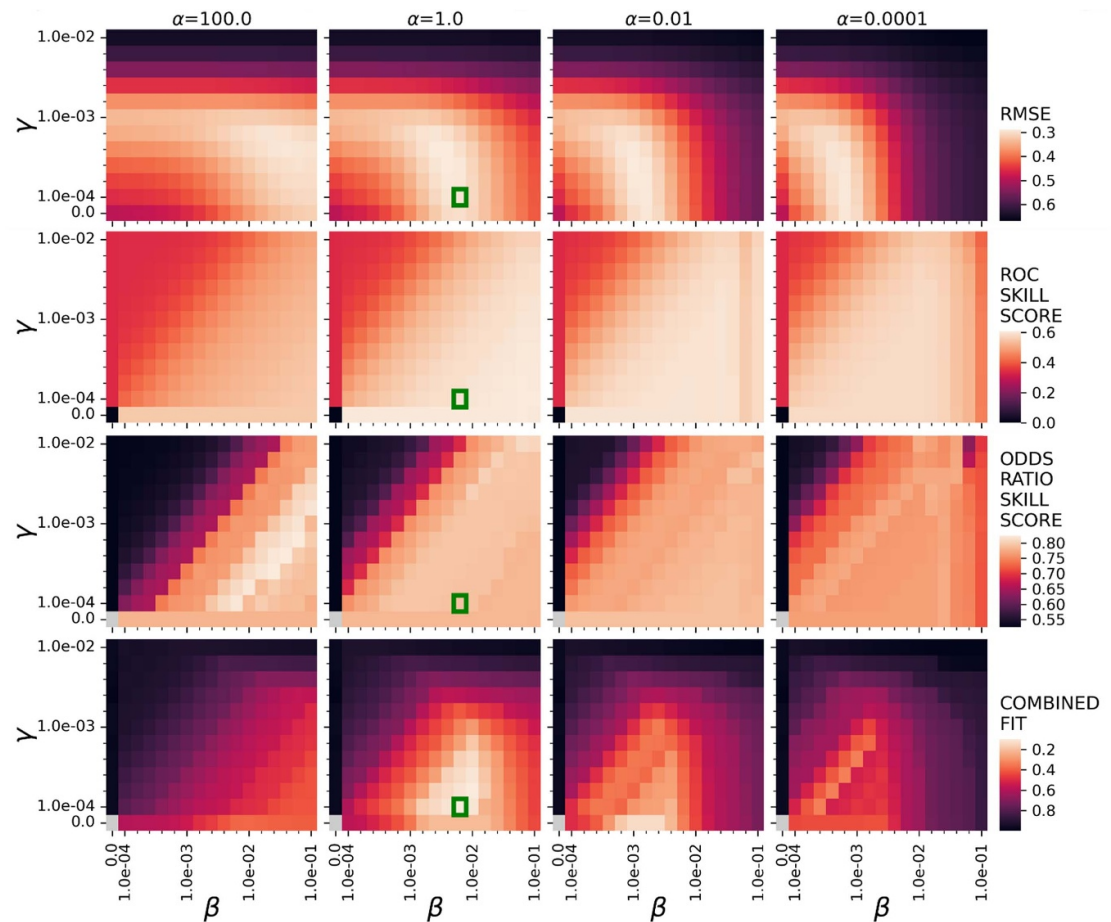


Figure 4. Metrics comparing simulations against observations across model parameter space for the training years 2015–2018. Definitions of α , β and γ are provided in table 1. Top row: root mean square error (RMSE) of zone averages; second row from top: ROC skill score (RSS) for model prediction at observation sites; third row: odds ratio skill score (ORSS) for dichotomised simulated infections at observation sites; bottom row: overall best fit from combination of RMSE, RSS and ORSS (E_{overall} as in equation (6)). Lighter shading indicates a better fit. The parameter combination with the best fit is outlined.

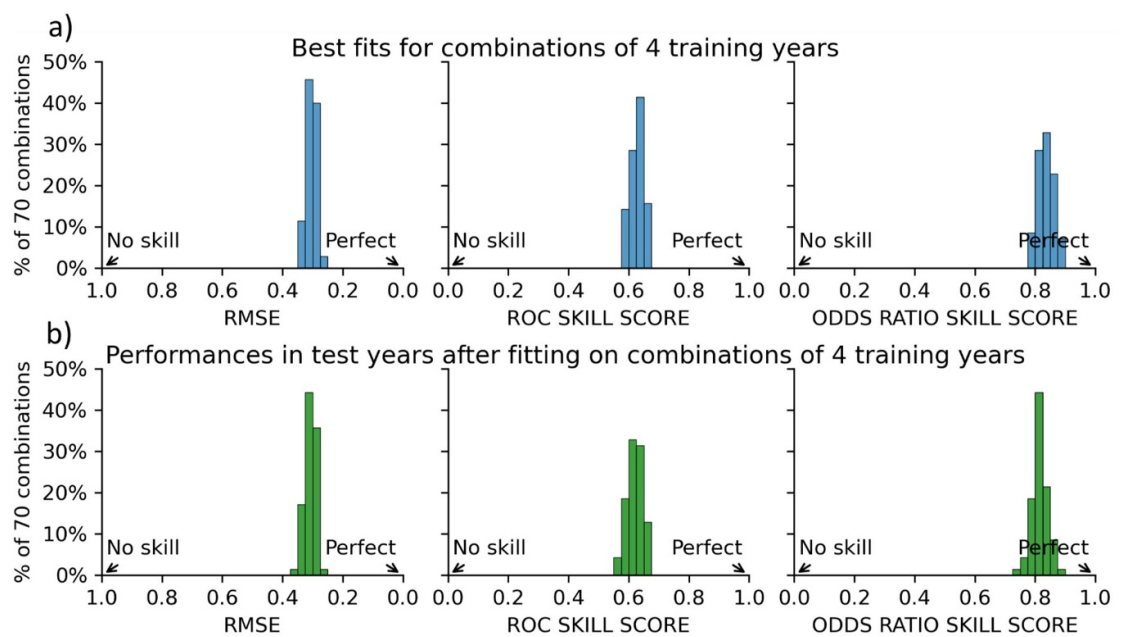
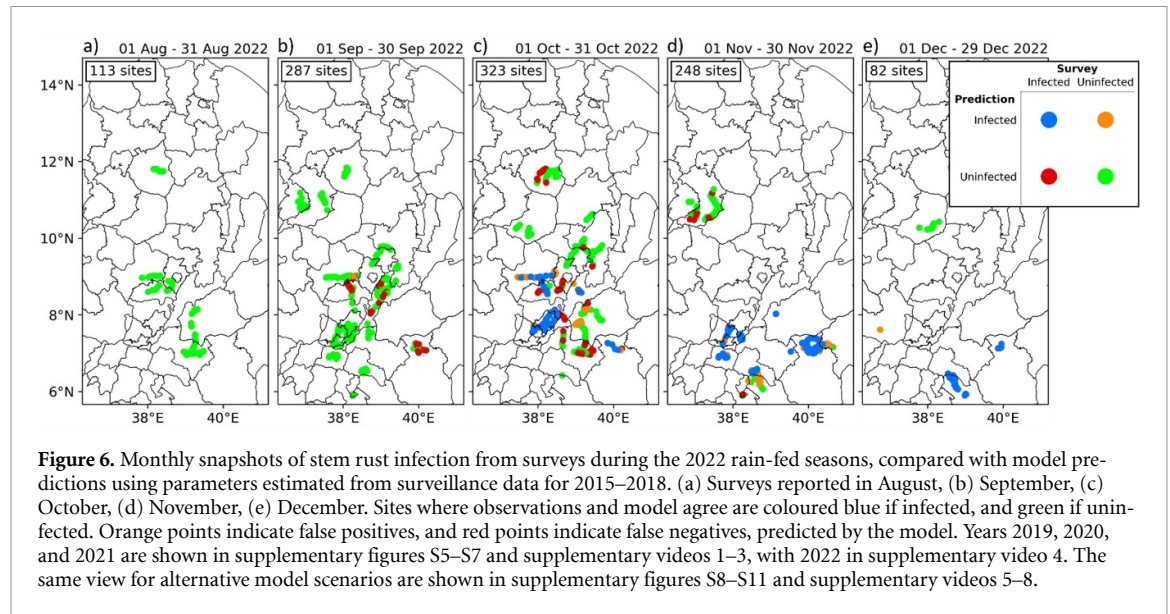


Figure 5. Cross-validation results: metrics for different combinations of four training years across the period 2015–2022. (a) Optimum fit in training years, (b) performance in corresponding test years.



of training period, spore deposition rates were negligible below 1 or 100 spores $\text{m}^{-2} [3 \text{ h}]^{-1}$ (α), and the role for spore deposition from background sources (γ) was weak or negligible. Values of RMSE were in the range 0.25–0.35, RSS 0.58–0.68, and ORSS 0.78–0.90 (figure 5(a)), indicating generally good fits.

3.2. Evaluation

The performance in test years is shown in figure 5(b). The model is validated by the metric values that are close to those found in the training period (figure 5(a)). A visual comparison of model predictions with observation is shown for rain-fed seasons during 2022 in figure 6. Stem rust levels and the rate of epidemic progress were moderate during 2022 compared with 2015–2021 (figure 2(c)). The model correctly predicted no infections in the beginning of the season in August (green points, figure 6(a)). The model was slow to predict the first sightings of stem rust infection (red points) in September (figure 6(b)), while capturing most of the disease spread in later months (blue points, figures 6(c) and (d)) and correctly predicting where infection was low or absent during the final months, November and December (blue and green points, figures 6(d) and (e)). In this case, the overall model accuracy was 0.82.

3.3. Evaluating alternative model scenarios

3.3.1. Model formulation

We compared the baseline model formulation with several simplified variants (figure 7). Our results indicate that dispersal from known sources of inoculum (referred to as ‘assimilation’ in figure 7) and long-distance dispersal have a strong influence on predictability. When the model was restricted so that surveys were not assimilated ($\beta = 0$ in equation (5)), the performance was substantially worse than the baseline model formulation (figure 7(a), scenario labelled ‘No assimilation’). In the example year

of 2022, the restricted model resolved little of the observed disease progression: it did not predict any infections before 1st October, and afterwards infection was predicted for almost all sites except in some of some of the south-easternmost zones (supplementary figure S8 and supplementary video S5). In this case, the model accuracy was 0.66.

Moreover, when surveys were assimilated but long-distance dispersal was prevented (replacing D_{NAME} with D_{SOURCE} in equation (5)), the performance in terms of model agreement with survey data was again worse than the baseline formulation (figure 7(a), scenario labelled ‘No long-distance dispersal’). In the case of 2022, this model scenario responded more reasonably than ‘no assimilation’ in central and south-eastern zones but still over-predicted infection in the northern observed areas (supplementary figure S9 and supplementary video S6) and the accuracy was 0.71.

We investigated two alternative formulations in relation to spore challenge rate at deposition sites. The baseline formulation applies a logarithmic spore challenge. One alternative represents equal competition between spores to infect host tissue (‘linear spore challenge’):

$$f_1(\mathbf{x}, t) = C(\mathbf{x}, t) E(\mathbf{x}, t) [\mu D_{\text{NAME}}(\mathbf{x}, t) + \nu P_1(\mathbf{x}, t)], \quad (8)$$

where μ and ν are modifiers of infected area per spore. Another alternative represents increased competition (and even saturation) at higher spore load, consistent with processes such as reduced availability of host tissue (‘Poisson spore challenge’):

$$f_2(\mathbf{x}, t) = \eta C(\mathbf{x}, t) (1 - \exp(-[\kappa D_{\text{NAME}}(\mathbf{x}, t) + \lambda P_2(\mathbf{x}, t)] E(\mathbf{x}, t))), \quad (9)$$

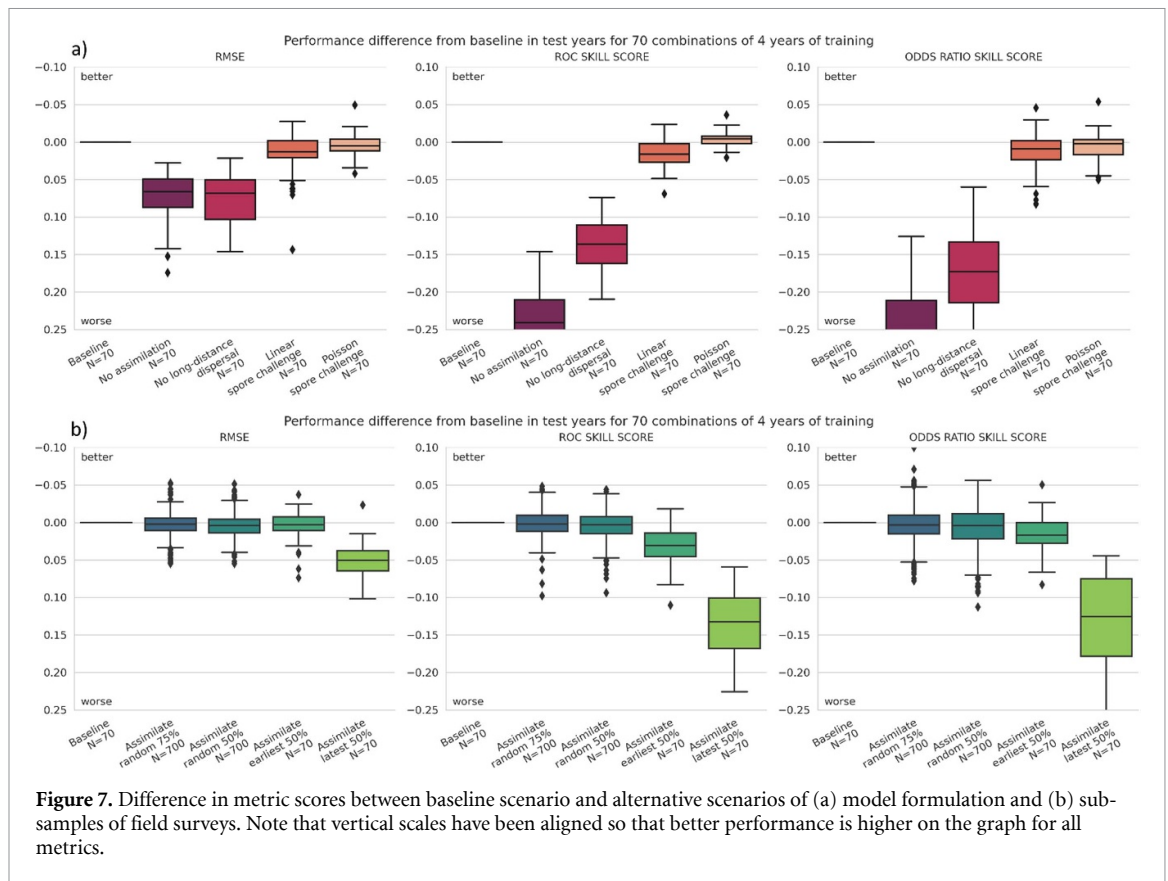


Figure 7. Difference in metric scores between baseline scenario and alternative scenarios of (a) model formulation and (b) subsamples of field surveys. Note that vertical scales have been aligned so that better performance is higher on the graph for all metrics.

where η is a modifier of infection rate and κ and λ are modifiers of infected area per spore. The comparison in figure 7(a) shows the scenario with linear spore challenge performed less well than the baseline whereas the scenario with Poisson spore challenge performed similarly.

We also considered neglecting environmental suitability from the integrated model. When setting $E = 1$, the performance values were similar to the baseline (see supplementary figure S12), indicating that the parameterised environmental conditions are generally satisfied when inoculum is present.

3.3.2. Survey subsampling

Field surveys of stem rust underpin the long-distance spore dispersal component of the integrated model. We evaluated the change in model performance with subsets of survey data compared with the baseline model with the full set of surveillance data (figure 7(b)). When the model was run with a random 50% or 75% of the survey data for each wheat-growing season (labelled in figure 7(b) as ‘Assimilate random 50%’ and ‘Assimilate random 75%’), and repeated with ten different random subsamples of each, the predictive performance of the model was only slightly reduced compared with the baseline.

When only the earliest 50% of survey data of each wheat-growing season were used in the model (‘Assimilate earliest 50%’), the performance was comparable with the baseline. However, when only the

latest 50% of annual surveys were available to the model (‘Assimilate latest 50%’), the performance was substantially degraded. This result is apparent in the example year 2022 (compare figure 6 with supplementary figures S10 and S11), when only the latest 50% of surveys were assimilated, the model predicts disease buildup in all areas by the end of the season and only a little heterogeneity mid-season (around October), whereas with only the earliest 50% of surveys assimilated, the associated dispersal rules out infection buildup in the northern surveyed zones and to some extent resolves spread to the south and east. Within the wheat-growing season, early-season observations appear to be influential in estimating sources of inoculum and initiating the crucial long-distance spore dispersal for the epidemic to spread to zones that were previously uninfected.

4. Discussion and conclusions

Our study has parameterised and validated a landscape-scale EWS to predict the spread of wheat stem rust in Ethiopia. The challenge of fitting and testing at large scale was overcome with the use of sparse field surveys taken across the main wheat producing areas in Ethiopia in combination with different metrics to constrain parameter space and to account for model performance at different scales. The difference in spatial scale between the field surveys and the model grid was accounted for by using

a combination of fitting metrics that function at different scales. Two metrics, ORSS and RSS focused on site-by-site (i.e. grid cell) comparisons, while RMSE assessed epidemic development at the larger landscape scale by comparing surveys and model predictions aggregated into administrative zones and successive time windows of two weeks (figure 2(c)). The site-by-site comparisons further distinguished between predicted presence/absence of infection (ORSS) and the predicted infected area fraction (RSS) in surveyed cells.

Fitting and validation of landscape-scale epidemiological models is challenging in practice, as it must be based on heterogeneous and sparse observations. Previous approaches (including Ellis *et al* 2025, Godding *et al* 2023, Meentemeyer *et al* 2011, Nguyen *et al* 2023) have used various statistical metrics including odds ratio, ROC area-under-curve, likelihoods, Moran's I (spatial autocorrelation), and mean squared difference at different spatial scales. Where epidemiological frameworks have accounted for long-distance dispersal with Eulerian or Eulerian–Lagrangian spore dispersal models, fitting and validation have taken different approaches. Some compare date of onset of predicted infection with past observations (climatological in the case of Aylor (2003), and for a specific year for Isard *et al* (2007)), some assess the timeseries of predicted spore concentration at a specific site in the domain (Newlands 2018), and Radici *et al* (2024) assess Cohen's kappa statistic (also known as Heidke skill score) in training and test years. The three metrics assessed in this paper are broadly analogous with the various metrics used in previous studies of landscape-scale epidemiological models, in that they are designed to assess the spatial and temporal agreement between model outputs and survey data, with, in the case of wheat rust, allowance for sparse test data in time and space.

The model performed well against test data, with comparatively high predictive scores for all three test metrics: RMSE values were in the range 0.25–0.35, RSS 0.58–0.68, and ORSS 0.78–0.90 (figure 5(b)). Because model fit and performance varied with the choice of training years, cross-validation was conducted by training and testing on every possible four-year combination within the eight-year study period. Cross-validation demonstrated that performance is sustained across the eight-year period despite year-to-year changes in surveillance coverage (cf figure 2), variety selection by growers, and pathogen strains within the landscape.

We evaluated how certain model components and the quantity and timing of surveillance data affected overall predictability of stem rust by comparing the model performance with some components of the model removed and also for different subsets of the

surveillance data (figure 7). Success in prediction was substantially enhanced by the assimilation of data from wheat rust surveys in near-real time, particularly in the use of observations early in the main wheat-growing season (figure 7(b)).

Our results also demonstrated the importance of accounting for long-distance spore dispersal in predicting epidemic spread of wheat stem rust across the heterogeneous wheat landscape (figure 7(a)). A novel feature of the integrated epidemiological model is the incorporation of a Lagrangian atmospheric dispersal model together with high resolution meteorological variables to track large numbers of particles (Meyer *et al* 2017a, Allen-Sader *et al* 2019, Gilligan 2024). The Lagrangian model accounts comprehensively for processes that affect aerial dispersal and deposition of spores. The processes include turbulence and wet and dry deposition that are missed by studies that focus on single-particle transport in the free troposphere (Radici *et al* 2022, 2023, 2024), or on Eulerian advection of spore concentration with coarse vertical resolution, as was used for the soybean rust aerobiological framework (Isard *et al* 2007). Estimates of the spread of wheat stem rust appear to be incomplete without fully accounting for long-distance aerial dispersal. While the underlying particle dispersion model has been validated in other contexts including volcanic ash, radionuclides, and chemical tracers (Webster *et al* 2012, Draxler *et al* 2015, Selvaratnam *et al* 2023), its validation for wheat rust spores has been limited to comparison with observed disease incidence (Olivera *et al* 2015, Meyer *et al* 2017a, Visser *et al* 2019). Spore trapping data over would serve to ground-truth and refine the spore dispersal model albeit requiring extensive coverage to match the scales of operation of the predictive models.

When used as part of an EWS, the stem rust forecast model offers practical advantages. Plant disease surveillance is valuable yet costly (Hershman *et al* 2011, Isard *et al* 2011). Linking surveillance with a predictive model to forecast risk as in the EWS, greatly increases the value of collecting surveillance data. In the case of the EWS for Ethiopia, we have shown that effective forecasts are possible, when less than 0.01% of wheat production area is surveyed. Risk assessment and interpretation of the forecasts does not require additional observations or data-gathering by small-holder farmers, for whom the system is designed.

The integrated epidemiological model can also be used to compare different scenarios to aid decision-making and guide effective use of limited resources. For example, model simulations can inform policy on risk from emerging virulent strains and changes in land management practices such as cultivation of resistant varieties, deployment of fungicides and expansion of irrigated wheat in Ethiopia (Effa *et al* 2023). This utility contrasts with statistical models which do

not readily enable scenario testing. The method to fit and validate the stem rust forecast model provides a basis for rapid testing of model developments and of revisions to input datasets. Pertinent future developments such as accounting at the landscape scale for host resistance to pathogen strains, and shifting norms of fungicide use, are challenged by the requirement for extensive and consistent supporting data.

In summary, the stem rust epidemiological model has been validated to predict infection outbreaks within a large region. The results demonstrated that long-distance dispersal has an important role in the spread of stem rust across the diverse agro-ecology of Ethiopia. The forecast system presented here can be adapted to other plant pathosystems where aerial dispersal is the primary risk of disease spread.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: <https://doi.org/10.5281/ZENODO.16039242> (Smith and Stutt 2025); <https://doi.org/10.5281/ZENODO.16689183> (Smith 2025).

Supplementary figures available at <http://doi.org/10.1088/1748-9326/ae2f70/data1>.

Supplementary video are listed below.

Predicting 2019 observations available at <http://doi.org/10.1088/1748-9326/ae2f70/data2>.

Predicting 2020 observations available at <http://doi.org/10.1088/1748-9326/ae2f70/data3>.

Predicting 2021 observations available at <http://doi.org/10.1088/1748-9326/ae2f70/data4>.

Predicting 2022 observations available at <http://doi.org/10.1088/1748-9326/ae2f70/data5>.

Prediction: no assimilation available at <http://doi.org/10.1088/1748-9326/ae2f70/data6>.

Prediction: no LDD available at <http://doi.org/10.1088/1748-9326/ae2f70/data7>.

Prediction: with early surveys available at <http://doi.org/10.1088/1748-9326/ae2f70/data8>.

Prediction: with late surveys available at <http://doi.org/10.1088/1748-9326/ae2f70/data9>.

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Conflicts of interest

The authors declare no known competing interests.

The study did not involve humans or animals.

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